

CI/CD Pipeline Guide for DevOps

Continuous Integration (CI) is the practice of automatically building and testing every code change as soon as it is merged. Continuous Delivery (CD) extends this by automatically deploying the validated build to staging or production environments.

Typical pipeline stages:

1. Source: the pipeline triggers on a git push or merge to the main branch.
2. Build: dependencies are installed and the application is compiled into an artifact such as a container image.
3. Test: unit tests, integration tests, and security scans run automatically.
4. Package: the tested artifact is pushed to a container registry like Docker Hub, Amazon ECR, or Google Artifact Registry.
5. Deploy: the artifact is deployed to a target environment using tools such as kubectl, Helm, or Terraform.
6. Verify: smoke tests and health checks confirm the deployment is healthy.

Popular CI/CD tools include GitHub Actions, GitLab CI, Jenkins, and Argo CD. A good pipeline fails fast on errors, keeps artifacts immutable, and requires minimal manual steps between commit and production.